SMART INDIA HACKATHON 2026

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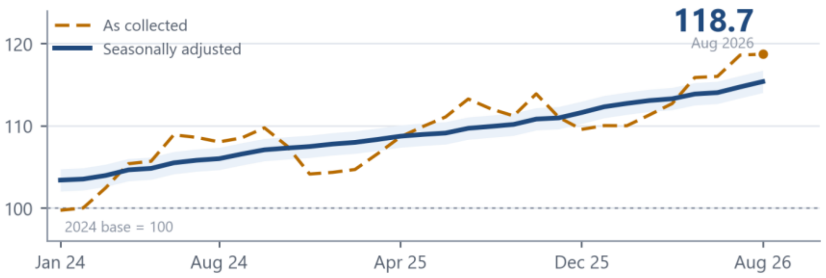
- **Problem Statement ID –** SIH26056
- **Problem Statement Title –** Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI)
- **Theme –** Travel & Tourism
- **PS Category –** Software
- **Team Name –** Bharat Bytes

VIMAAN Validated Index of Market Airfares for Augmenting National-statistics

Skyscanner tells you what a ticket costs. VIMAAN tells the RBI whether to raise rates.



Source: Modelled on the VIMAAN sampling design. Base 2024 = 100, as in CPI 2024.

517
ROUTES, 3× A DAY

78 → 517
VS DGCA TODAY

SDMX
FEED INTO CPI 2024

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PROPOSED SOLUTION

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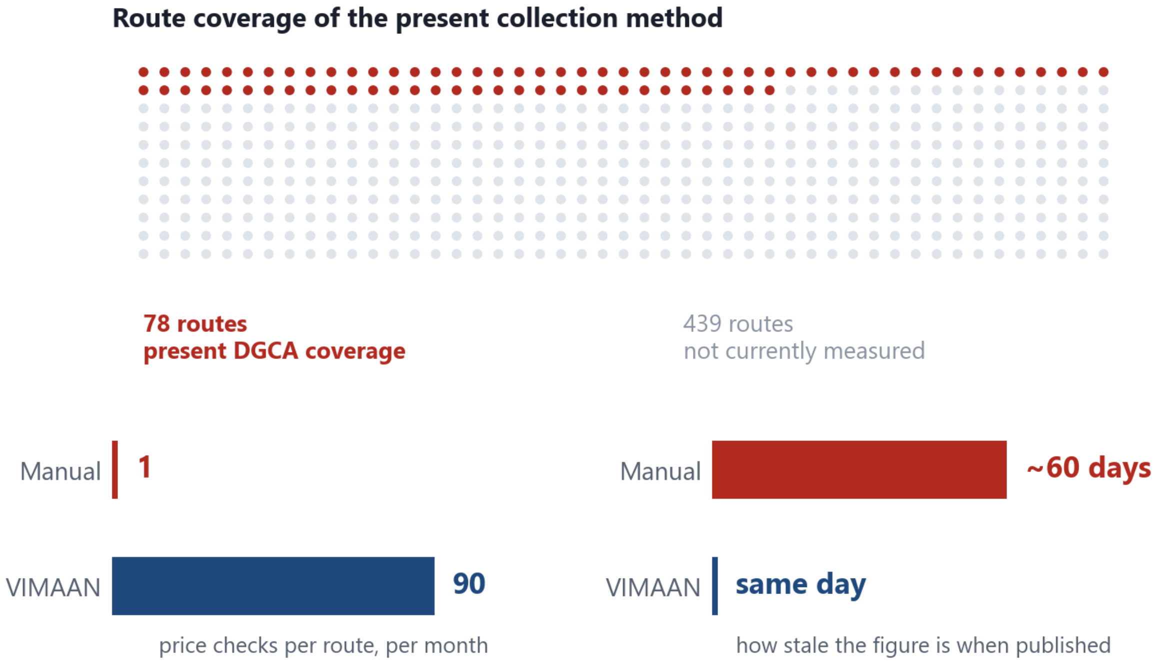
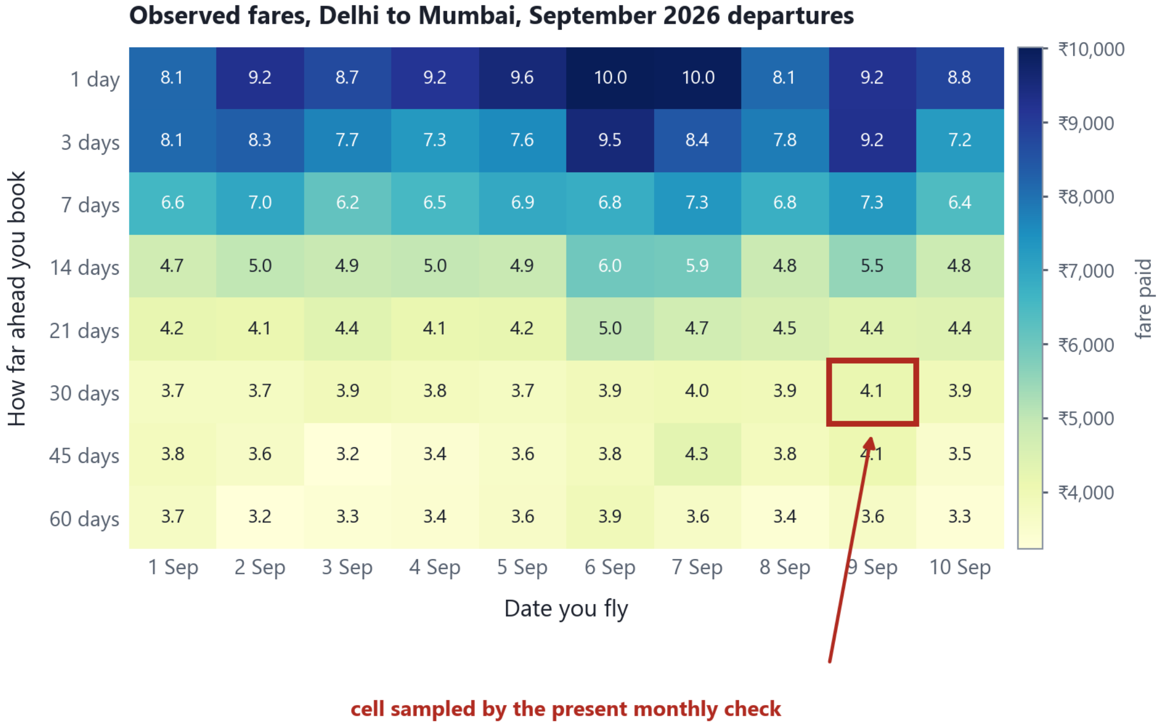
VIMAAN **Validated Index of Market Airfares for Augmenting National-statistics.** An automated replacement for the manual airfare price collection that currently supplies the CPI transport group.

EXHIBIT 2 DETAILED EXPLANATION OF THE PROPOSED SOLUTION



Source: Route count from DGCA Monthly Domestic Traffic Statistics. Present coverage per PIB PRID 1810467.

EXHIBIT 3 HOW IT ADDRESSES THE PROBLEM



Source: Fare grid modelled on the VIMAAN sampling design, Delhi–Mumbai, September 2026 departures.

INNOVATION AND UNIQUENESS OF THE SOLUTION

01 Full price surface sampled

Eight advance-purchase windows across every departure date in a 60-day horizon, each fare attributed to its month of departure.

02 Multilateral index estimators

Jevons elementary aggregates, then GEKS-Jevons on a 25-month rolling window, which removes chain drift.

03 Statutory collection basis

Rule 135(2) tariff pages and licensed APIs supply two of three lanes. robots.txt observed; no personal data.

04 Reproducible output

Hashed raw snapshots, a methodology endpoint for each value, and SDMX-JSON for direct ingestion by MoSPI.

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TECHNICAL APPROACH

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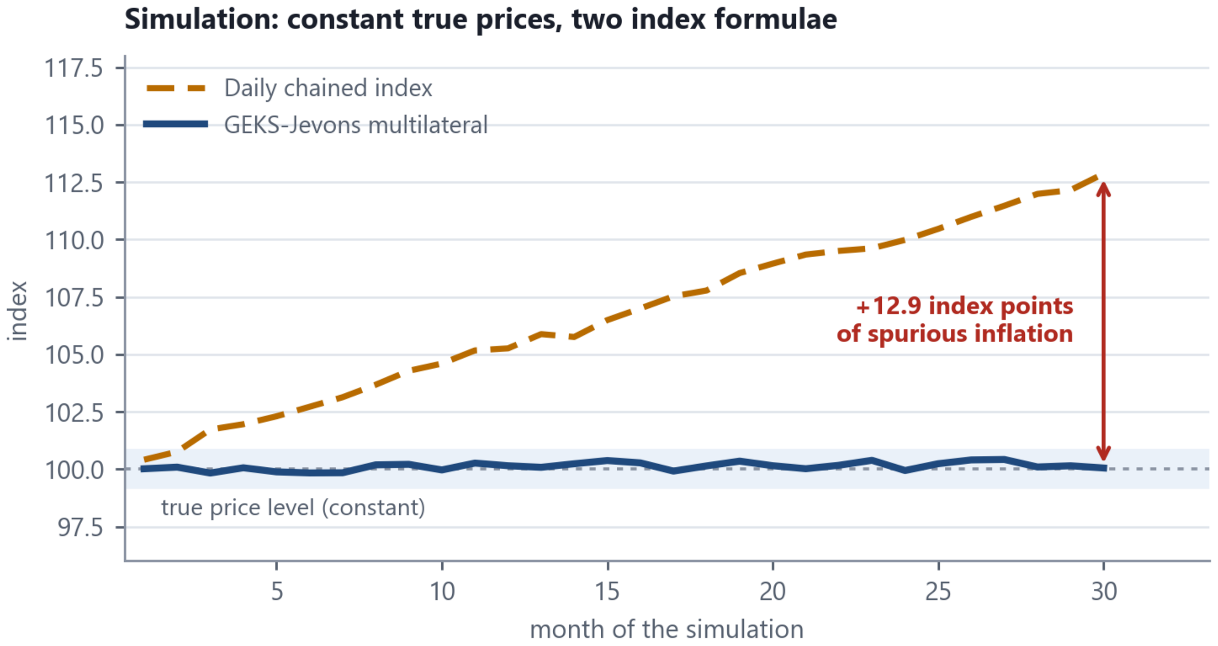
METHODOLOGY AND PROCESS FOR IMPLEMENTATION

1 COLLECT	2 CLEAN	3 ADJUST	4 INDEX	5 PUBLISH
statutory tariff pages, licensed APIs, public portals	IATA codes, one currency, all-in fares, outliers out	hedonic model strips stops, duration, cabin, baggage	Jevons → GEKS–Jevons → DGCA passenger weights	REST + SDMX-JSON, dashboard, surge alerts to DGCA

The pipeline runs without manual intervention. Sampling parameters are given in the specification, right.

EXHIBIT 4 **EFFECT OF THE INDEX FORMULA ON THE PUBLISHED FIGURE**



Source: Simulation with constant true prices. Estimators per ONS Methodology Working Paper 12 and the ILO/IMF CPI Manual 2020.

P_J(0,t) = $\Pi_j (p_j^t / p_j^o)$^(1/n)   Index(t) = $\Sigma_i w_i I_i(t) / \Sigma_i w_i \times 100$, w_i = DGCA base-period passengers

TECHNOLOGIES TO BE USED

COLLECT Python 3.11 · Playwright · httpx · Redis + RQ

STORE PostgreSQL 16 + TimescaleDB · Parquet / DuckDB

PROCESS Polars · NumPy / SciPy · Great Expectations

STATISTICS statsmodels · X-13ARIMA-SEATS · index library

SERVE FastAPI · Pydantic v2 · SDMX-JSON · Swagger

INTERFACE React 18 + Vite · visx · MapLibre + deck.gl

OPERATE Docker Compose · Prometheus + Grafana · CI

INDEX SPECIFICATION

Collection	3 times daily, 02:00 / 11:00 / 19:00 IST
Lead buckets	L1, L3, L7, L14, L21, L30, L45, L60
Horizon	60 days forward, every departure date
Attribution	month of departure (acquisition basis in parallel)
Elementary	Jevons geometric mean, per route and bucket
Multilateral	GEKS-Jevons, cross-checked with Time-Product-Dummy
Window	25 months, rolling, extended by mean splicing
Weights	DGCA base-period route passengers
Seasonal	X-13ARIMA-SEATS, Indian moving-holiday regressors
Output	REST, SDMX-JSON, monthly bulletin

OUT OF SCOPE BY DESIGN

No CAPTCHA solving, no login or paywall circumvention, no fingerprint spoofing, and no personal data at any stage. Two of the three intake lanes do not scrape.

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FEASIBILITY AND VIABILITY

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ANALYSIS OF THE FEASIBILITY OF THE IDEA

COLLECTION BASIS IS STATUTORY

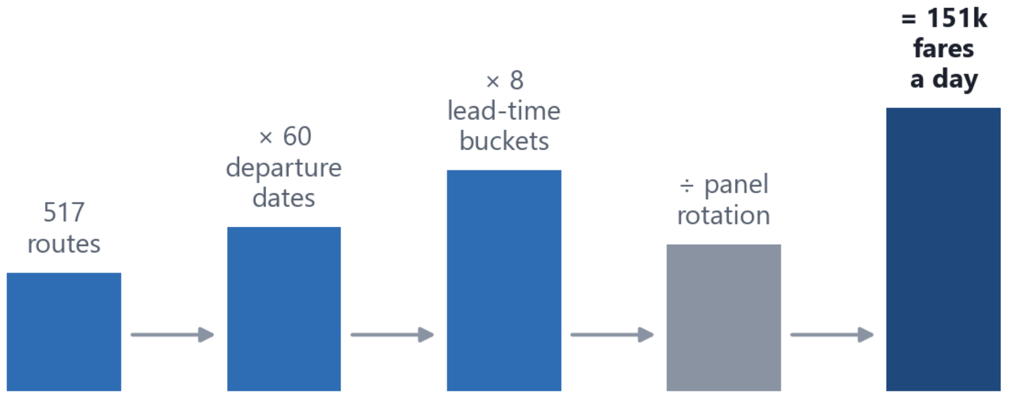
Rule 135(2), Aircraft Rules 1937 obliges every airline to publish its tariff. Two of our three intake lanes therefore need no scraping at all.

COMPONENTS ARE STANDARD AND UNLICENSED

Python, Playwright, PostgreSQL, statsmodels, FastAPI, React, Docker. Nothing exotic, nothing licensed, nothing to procure.

EXHIBIT 5 IMPLIED DATA VOLUME

Daily observation count implied by the sampling design

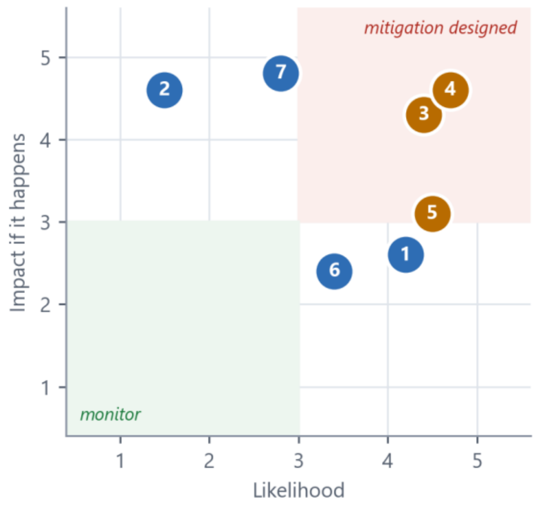


Panel rotation keeps the daily load within a single server.

Source: Derived from the sampling design set out in the specification on slide 3.

EXHIBIT 6 POTENTIAL CHALLENGES AND RISKS

Risk register: likelihood against impact



Mitigations are listed under Strategies

1. Portal blocks the collector
2. Legal challenge to collection
3. No 2024 base-period fares
4. Chain drift inflates the index
5. Listed fare is not fare paid
6. Site layout breaks the parser
7. Demo-day network failure

Source: Team assessment. Each entry is answered in the strategies below.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

1	**Portal blocks us**	three independent lanes; A and B are statutory and contractual
2	**Legal challenge**	Rule 135(2) mandate + Collection of Statistics Act 2008; zero PII
3	**No 2024 base fares**	backcast from DGCA yields; published provisional with a band
4	**Chain drift**	GEKS–Jevons and TPD on a 25-month rolling window, mean spliced
5	**Listed ≠ paid**	yield calibration from DGCA route revenue, published with uncertainty
6	**Parser breaks**	config-driven selectors, schema-drift detector, per-lane SLO alarms
7	**Demo network dies**	seeded 21-day dataset, fully offline Docker Compose, backup video

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IMPACT AND BENEFITS

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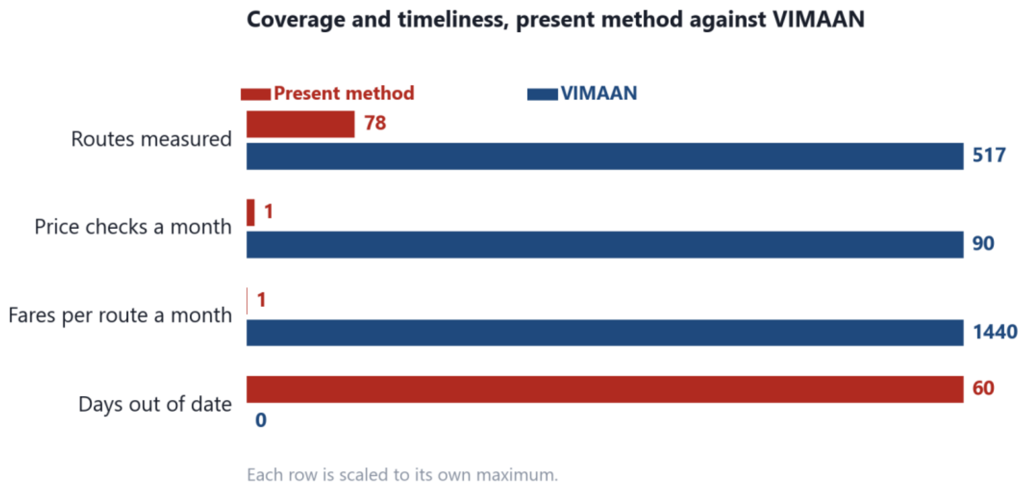
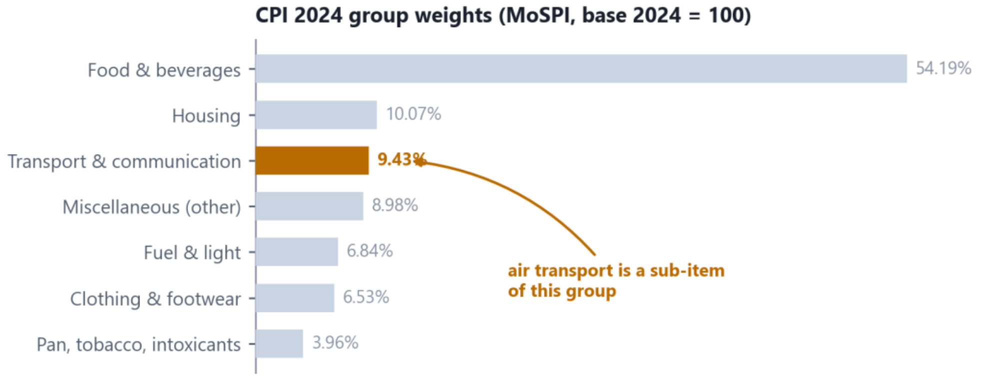
EXHIBIT 7 POTENTIAL IMPACT ON THE TARGET AUDIENCE



Source: Present method per PIB PRID 1810467. VIMAAN figures are design parameters.

EXHIBIT 8 POSITION WITHIN THE CPI 2024 BASKET



Source: MoSPI, Consumer Price Index base 2024 = 100. First release, PIB PRID 2227012.

TRANSMISSION OF THE INDEX INTO POLICY

Millions of fares → **VIMAAN index** → **CPI 2024** → **Repo rate**

Millions of fares	VIMAAN index	CPI 2024	Repo rate
517 routes, 3× daily	quality-adjusted	transport, 9.43%	EMIs for 140 crore

Source: CPI group weight from MoSPI. Repo transmission is the standard monetary policy channel, not a claim specific to this project.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL)

ECONOMIC — MOSPI AND THE RBI

An unbiased, quality-adjusted airfare component feeds headline CPI, which sets the repo rate. Better input, better monetary policy.

ADMINISTRATIVE — DGCA

Tariff monitoring goes from 78 hand-checked routes a month to 517 routes three times a day, with automatic fare-surge alerts.

SOCIAL — CITIZENS AND TRAVELLERS

Public route and lead-time indices expose the true advance-purchase penalty and festival-season surges. No such statistic exists for India today.

ENVIRONMENTAL — MOCA

Field price collection is replaced by an automated feed, removing enumerator travel. The ATF pass-through model shows how much of a fare rise is fuel.

STATISTICAL — MOSPI, RBI AND THE IMF

SDMX-JSON is ingested without manual handling. Hashed provenance makes every published value defensible under statistical audit and under RTI.

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RESEARCH AND REFERENCES

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DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK · EACH SOURCE, AND WHAT IT SETTLES

A INDIAN STATUTORY BASIS

Establishes that the data is public and that MoSPI may collect it

1. **MoSPI — FAQs on the CPI 2024 Series, Annexure V**
   airfare prices are collected through online platforms **mospi.gov.in**
2. **PIB — first CPI release on base 2024 = 100, 12 Feb 2026**
   358 items · COICOP 2018 · Transport weight 9.43% **PRID 2227012**
3. **Aircraft Rules, 1937 — Rule 135(2)**
   every airline shall publish its established tariff **civilaviation.gov.in**
4. **Collection of Statistics Act, 2008**
   statutory mandate for official price collection **indiacode.nic.in**
5. **PIB / MoCA — DGCA Tariff Monitoring Unit**
   78 routes, monthly, read by hand from airline sites **PRID 1810467**
6. **DGCA — Monthly Domestic Traffic Statistics**
   route-wise passengers — our index weights **dgca.gov.in**
7. **MoSPI — Expert Group Report on updation of the CPI**
   weight derivation and item-basket methodology **mospi.gov.in**

B INTERNATIONAL METHOD

Establishes that our estimators are the ones NSOs already use

8. **ONS — aggregate index of air fares (FOI-2023-1164)**
   fixed advance-purchase windows; month-of-departure basis **ons.gov.uk**
9. **ONS — Methodology Working Paper 12**
   index-number methods on web-scraped price data **ons.gov.uk**
10. **ONS — multilateral methods in consumer price statistics**
    GEKS–Jevons, TPD, rolling windows, splicing **ons.gov.uk**
11. **ILO / IMF / OECD / Eurostat — CPI Manual 2020**
    why Jevons, not Carli or Dutot, for elementary aggregates **imf.org**
12. **Statistics Canada — Air Transportation Index, 2020**
    dynamic airline pricing inside an official index **statcan.gc.ca**
13. **Statistics Netherlands / Eurostat — web scraping**
    25-month window guidance and chain-drift evidence **unece.org**
14. **IOCL — Aviation Turbine Fuel price circulars**
    input to the fuel pass-through model **iocl.com**

STANDING OF THE METHOD

The collection basis is statutory and the estimators are those already in use by the UK Office for National Statistics, Statistics Canada and Statistics Netherlands on web-collected prices. The contribution here is their application to Indian route data, with the supporting evidence published alongside every figure.

C OUR OWN VALIDATION

Establishes how anyone can check the number we publish

DATASETS ASSEMBLED BEFORE HOUR ZERO

routes_top500.csv (DGCA), dgca_pax_weights.csv, base_2024_fares.csv (backcast), airports.geojson, atf_prices.csv (IOCL), festival_calendar.csv

HOW THE INDEX WILL BE VALIDATED

Benchmarked against MoSPI's own manually collected airfare series over overlapping months. Cross-lane divergence tests with a flag threshold. Leave-one-route-out sensitivity on the DGCA weights. A published revision policy.

DISTINCTION FROM A FARE-SEARCH PRODUCT

Unit of observation	a price surface, not a single price
Elementary aggregate	Jevons geometric mean, not an arithmetic average
Multilateral stage	GEKS-Jevons on a rolling window
Attribution	month of departure, DGCA passenger weights
Delivery	an SDMX statistical feed, not only a dashboard
Auditability	a reproducibility hash on every published value
Collection basis	two of three lanes statutory, not scraped